M2 Conditional probability
A level only

Name:

Class:

Date:

Time: **483 minutes**

Marks: **398 marks**

Comments:

Q1.

A teacher in a college asks her mathematics students what other subjects they are studying.

She finds that, of her 24 students:

12 study physics
8 study geography
4 study geography and physics

(a) A student is chosen at random from the class.

Determine whether the event 'the student studies physics' and the event 'the student studies geography' are independent.

(2)

(b) It is known that for the whole college:

the probability of a student studying mathematics is $\frac{1}{5}$

the probability of a student studying biology is $\frac{1}{6}$

the probability of a student studying biology given that they study mathematics is $\frac{3}{8}$

Calculate the probability that a student studies mathematics or biology or both.

(4)

(Total 6 marks)

Q2.

A sample of 200 households was obtained from a small town.

Each household was asked to complete a questionnaire about their purchases of takeaway food.
A is the event that a household regularly purchases Indian takeaway food.
B is the event that a household regularly purchases Chinese takeaway food.

It was observed that $\mathrm{P}(B|A) = 0.25$ and $\mathrm{P}(A|B) = 0.1$
Of these households, 122 indicated that they did **not** regularly purchase Indian or Chinese takeaway food.
A household is selected at random from those in the sample.
Find the probability that the household regularly purchases **both** Indian and Chinese takeaway food.

(Total 6 marks)

Q3.

Roger is an active retired lecturer. Each day after breakfast, he decides whether the weather for that day is going to be fine (F), dull (D) or wet (W). He then decides on only one of four activities for the day: cycling (C), gardening (G), shopping (S) or relaxing (R). His decisions from day to day may be assumed to be independent.

The table shows Roger's probabilities for each combination of weather and activity.

		Weather		
		Fine (F)	**Dull (D)**	**Wet (W)**
Activity	**Cycling (C)**	0.30	0.10	0
	Gardening (G)	0.25	0.05	0
	Shopping (S)	0	0.10	0.05
	Relaxing (R)	0	0.05	0.10

(a) Find the probability that, on a particular day, Roger decided:

(i) that it was going to be fine and that he would go cycling;

(ii) on either gardening or shopping;

(iii) to go cycling, given that he had decided that it was going to be fine;

(iv) that it was going to be fine, given that he did **not** go cycling.

(7)

(b) Calculate the probability that, on a particular Saturday and Sunday, Roger decided that it was going to be fine and decided on the same activity for both days.

(3)

(Total 10 marks)

Q4.

Roger is an active retired lecturer. Each day after breakfast, he decides whether the weather for that day is going to be fine (F), dull (D) or wet (W). He then decides on only one of four activities for the day: cycling (C), gardening (G), shopping (S) or relaxing (R). His decisions from day to day may be assumed to be independent.

The table shows Roger's probabilities for each combination of weather and activity.

		Weather		
		Fine (F)	**Dull (D)**	**Wet (W)**

Activity	**Cycling (C)**	0.30	0.10	0
	Gardening (G)	0.25	0.05	0
	Shopping (S)	0	0.10	0.05
	Relaxing (R)	0	0.05	0.10

(a) Find the probability that, on a particular day, Roger decided:

(i) that it was going to be fine and that he would go cycling;

(ii) on either gardening or shopping;

(iii) to go cycling, given that he had decided that it was going to be fine;

(iv) **not** to relax, given that he had decided that it was going to be dull;

(v) that it was going to be fine, given that he did **not** go cycling.

(9)

(b) Calculate the probability that, on a particular Saturday and Sunday, Roger decided that it was going to be fine and decided on the same activity for both days.

(3)

(Total 12 marks)

Q5.

Alison is a member of a tenpin bowling club which meets at a bowling alley on Wednesday and Thursday evenings.

The probability that she bowls on a Wednesday evening is 0.90. Independently, the probability that she bowls on a Thursday evening is 0.95.

(a) Calculate the probability that, during a particular week, Alison bowls on:

(i) two evenings;

(ii) exactly one evening.

(3)

(b) David, a friend of Alison, is a member of the same club.

The probability that he bowls on a Wednesday evening, given that Alison bowls on that evening, is 0.80. The probability that he bowls on a Wednesday evening, given that Alison does not bowl on that evening, is 0.15.

The probability that he bowls on a Thursday evening, given that Alison bowls on that evening, is 1. The probability that he bowls on a Thursday evening, given that Alison does not bowl on that evening, is 0.

Calculate the probability that, during a particular week:

(i) Alison and David bowl on a Wednesday evening;

(2)

(ii) Alison and David bowl on both evenings;

(2)

(iii) Alison, but not David, bowls on a Thursday evening;

(1)

(iv) neither bowls on either evening.

(3)
(Total 11 marks)

Q6.

On a rail route between two stations, A and B, 90% of trains leave A on time and 10% of trains leave A late.

Of those trains that leave A on time, 15% arrive at B early, 75% arrive on time and 10% arrive late.

Of those trains that leave A late, 35% arrive at B on time and 65% arrive late.

(a) Represent this information by a fully-labelled tree diagram.

(3)

(b) Hence, or otherwise, calculate the probability that a train:

(i) arrives at B early or on time;

(ii) left A on time, given that it arrived at B on time;

(iii) left A late, given that it was not late in arriving at B.

(7)

(c) Two trains arrive late at B. Assuming that their journey times are independent, calculate the probability that exactly one train left A on time.

(4)
(Total 14 marks)

Q7.

Twins Alec and Eric are members of the same local cricket club and play for the club's under 18 team.

The probability that Alec is selected to play in any particular game is 0.85.
The probability that Eric is selected to play in any particular game is 0.60.
The probability that both Alec and Eric are selected to play in any particular game is 0.55.

(a) By using a table, or otherwise:

(i) show that the probability that neither twin is selected for a particular game is 0.10;

(ii) find the probability that at least one of the twins is selected for a particular game;

(iii) find the probability that exactly one of the twins is selected for a particular game.

(5)

(b) The probability that the twins' younger brother, Cedric, is selected for a particular game is:

0.30 given that both of the twins have been selected;
0.75 given that exactly one of the twins has been selected;
0.40 given that neither of the twins has been selected.

Calculate the probability that, for a particular game:

(i) all three brothers are selected;

(ii) at least two of the three brothers are selected.

(6)
(Total 11 marks)

Q8.

A survey of the 640 properties on an estate was undertaken. Part of the information collected related to the number of bedrooms and the number of toilets in each property.

This information is shown in the table.

		Number of toilets				
		1	**2**	**3**	**4 or more**	**Total**
Number of bedrooms	**1**	46	14	0	0	60
	2	24	67	23	0	114
	3	7	72	99	16	194
	4	0	19	123	48	190
	5 or more	0	0	11	71	82
	Total	77	172	256	135	640

(a) A property on the estate is selected at random.

Find, giving your answer to three decimal places, the probability that the property has:

(i) exactly 3 bedrooms;

(1)

(ii) at least 2 toilets;

(2)

(iii) exactly 3 bedrooms and at least 2 toilets;

(2)

(iv) at most 3 bedrooms, given that it has exactly 2 toilets.

(3)

(b) Three properties are selected at random from those on the estate which have exactly 3 bedrooms.

Calculate the probability that one property has 2 toilets, one has 3 toilets and the other has at least 4 toilets. Give your answer to three decimal places.

(4)

(Total 12 marks)

Q9.

A survey of the 640 properties on an estate was undertaken. Part of the information collected related to the number of bedrooms and the number of toilets in each property.

This information is shown in the table.

		Number of toilets				
		1	**2**	**3**	**4 or more**	**Total**
Number of bedrooms	**1**	46	14	0	0	60
	2	24	67	23	0	114
	3	7	72	99	16	194
	4	0	19	123	48	190
	5 or more	0	0	11	71	82
	Total	77	172	256	135	640

(a) A property on the estate is selected at random.

Find, giving your answer to three decimal places, the probability that the property has:

(i) exactly 3 bedrooms;

(1)

(ii) at least 2 toilets;

(2)

(iii) exactly 3 bedrooms and at least 2 toilets;

(2)

(iv) at most 3 bedrooms, given that it has exactly 2 toilets.

(3)

(b) Use relevant answers from part (a) to show that the number of toilets is **not** independent of the number of bedrooms.

(2)

(c) Three properties are selected at random from those on the estate which have exactly 3 bedrooms.

Calculate the probability that one property has 2 toilets, one has 3 toilets and the other has at least 4 toilets. Give your answer to three decimal places.

(4)
(Total 14 marks)

Q10.

A hotel has three types of room: double, twin and suite. The **percentage** of rooms in the hotel of each type is 40, 45 and 15 respectively.

Each room in the hotel may be occupied by 0, 1, 2, or 3 or more people.

The **proportional** occupancy of **each** type of room is shown in the table.

		Occupancy			
		0	**1**	**2**	**3 or more**
Room	**Double**	0.15	0.35	0.45	0.05
	Twin	0.05	0.55	0.30	0.10
	Suite	0.10	0.20	0.55	0.15

For example, the probability that, on a particular night, a double room has exactly 2 occupants is 0.45.

On a particular night, a room is selected at random. Find the probability that this room is:

(a) an unoccupied suite;

(1)

(b) occupied by 2 or more people;

(2)

(c) unoccupied;

(2)

(d) a double room, given that it is unoccupied;

(2)

(e) a suite, given that it is occupied.

(3)
(Total 10 marks)

Q11.

The number of MPs in the House of Commons was 645 at the beginning of August 2009. The genders of these MPs and the political parties to which they belonged are shown in the table.

		Political Party				
		Labour	**Conservative**	**Liberal Democrat**	**Other**	**Total**
Gender	**Male**	255	175	54	35	519
	Female	94	18	9	5	126
	Total	349	193	63	40	645

(a) One MP was selected at random for an interview. Calculate, to three decimal places, the probability that the MP was:

(i) a male Conservative;

(1)

(ii) a male;

(1)

(iii) a Liberal Democrat;

(1)

(iv) Labour, given that the MP was female.

(2)

(b) Two female MPs were selected at random for an enquiry. Calculate, to three decimal places, the probability that both MPs were Labour.

(2)

(c) Three MPs were selected at random for a committee. Calculate, to three decimal places, the probability that exactly one MP was Labour and exactly one MP was Conservative.

(4)
(Total 11 marks)

Q12.

The number of MPs in the House of Commons was 645 at the beginning of August 2009.

The genders of these MPs and the political parties to which they belonged are shown in the table.

		Political Party				
		Labour	Conservative	Liberal Democrat	Other	Total
Gender	Male	255	175	54	35	519
	Female	94	18	9	5	126
	Total	349	193	63	40	645

(a) One MP was selected at random for an interview. Calculate, to three decimal places, the probability that the MP was:

(i) a male Conservative;

(1)

(ii) a male;

(1)

(iii) a Liberal Democrat;

(1)

(iv) Labour, given that the MP was female;

(2)

(v) male, given that the MP was not Labour.

(3)

(b) Two female MPs were selected at random for an enquiry. Calculate, to three decimal places, the probability that both MPs were Labour.

(2)

(c) Three MPs were selected at random for a committee. Calculate, to three decimal places, the probability that exactly one MP was Labour and exactly one MP was Conservative.

(4)
(Total 14 marks)

Q13.

Clay pigeon shooting is the sport of shooting at special flying clay targets with a shotgun.

(a) Rhys, a novice, uses a single-barrelled shotgun. The probability that he hits a target is 0.45, and may be assumed to be independent from target to target.

Determine the probability that, in a series of shots at 15 targets, he hits:

(i) at most 5 targets;

(1)

(ii) more than 10 targets;

(2)

(iii) exactly 6 targets;

(2)

(iv) at least 5 but at most 10 targets.

(3)

(b) Sasha, an expert, uses a double-barrelled shotgun. She shoots at each target with the gun's first barrel and, only if she misses, does she then shoot at the target with the gun's second barrel.

The probability that she hits a target with a shot using her gun's first barrel is 0.85. The conditional probability that she hits a target with a shot using her gun's second barrel, given that she has missed the target with a shot using her gun's first barrel, is 0.80. Assume that Sasha's shooting is independent from target to target.

(i) Show that the probability that Sasha hits a target is 0.97.

(2)

(ii) Determine the probability that, in a series of shots at 50 targets, Sasha hits at least 48 targets.

(3)

(iii) In a series of shots at 80 targets, calculate the mean number of times that Sasha shoots at targets with her gun's second barrel.

(2)
(Total 15 marks)

Q14.

Emma visits her local supermarket every Thursday to do her weekly shopping.

The event that she buys orange juice is denoted by J, and the event that she buys bottled water is denoted by W. At each visit, Emma may buy neither, or one, or both of these items.

(a) Complete the table of probabilities, printed below, for these events, where J' and W' denote the events 'not J' and 'not W' respectively.

(3)

(b) Hence, or otherwise, find the probability that, on any given Thursday, Emma buys either orange juice or bottled water but not both.

(2)

(c) Show that:

(i) the events J and W are **not** mutually exclusive;

(ii) the events J and W are **not** independent.

	J	J'	Total
W			0.65
W'	0.15		
Total		0.30	1.00

(3)
(Total 8 marks)

Q15.

(a) Emma visits her local supermarket every Thursday to do her weekly shopping.

The event that she buys orange juice is denoted by J, and the event that she buys bottled water is denoted by W. At each visit, Emma may buy neither, or one, or both of these items.

(i) Complete the table of probabilities, printed below, for these events, where J' and W' denote the events 'not J' and 'not W' respectively.

(3)

(ii) Hence, or otherwise, find the probability that, on any given Thursday, Emma buys either orange juice or bottled water but not both.

(2)

(iii) Show that:

(**A**) the events J and W are **not** mutually exclusive;

(**B**) the events J and W are **not** independent.

(3)

(b) Rhys visits the supermarket every Saturday to do his weekly shopping. Items that he may buy are milk, cheese and yogurt.

The probability, P(M), that he buys milk on any given Saturday is 0.85.
The probability, P(C), that he buys cheese on any given Saturday is 0.60.
The probability, P(Y), that he buys yogurt on any given Saturday is 0.55.
The events M, C and Y may be assumed to be independent.

Calculate the probability that, on any given Saturday, Rhys buys:

(i) none of the 3 items;

(2)

(ii) exactly 2 of the 3 items.

	J	J'	Total
W			0.65

W'	0.15		
Total		0.30	1.00

(3)
(Total 13 marks)

Q16.

An IT help desk has three telephone stations: Alpha, Beta and Gamma. Each of these stations deals only with telephone enquiries.

The probability that an enquiry is received at Alpha is 0.60.
The probability that an enquiry is received at Beta is 0.25.
The probability that an enquiry is received at Gamma is 0.15.

Each enquiry is resolved at the station that receives the enquiry. The **percentages** of enquiries resolved within various times at **each** station are shown in the table.

		Time		
		≤ 1 hour	**≤ 24 hours**	**≤ 72 hours**
Station	**Alpha**	55	80	100
	Beta	60	85	100
	Gamma	40	75	100

For example:
80 per cent of enquiries received at Alpha are resolved within 24 hours;
25 per cent of enquiries received at Alpha take between 1 hour and 24 hours to resolve.

(a) Find the probability that an enquiry, selected at random, is:

(i) resolved at Gamma;

(1)

(ii) resolved at Alpha within 1 hour;

(1)

(iii) resolved within 24 hours;

(2)

(iv) received at Beta, given that it is resolved within 24 hours.

(3)

(b) A random sample of 3 enquiries was selected.

Given that all 3 enquiries were resolved within 24 hours, calculate the probability that they were all received at:

(i) Beta;

(2)

(ii) the same station.

(4)
(Total 13 marks)

Q17.

Each school-day morning, three students, Rita, Said and Ting, travel independently from their homes to the same school by one of three methods: walk, cycle or bus. The table shows the probabilities of their independent daily choices.

	Walk	Cycle	Bus
Rita	0.65	0.10	0.25
Said	0.40	0.45	0.15
Ting	0.25	0.55	0.20

(a) Calculate the probability that, on any given school-day morning:

(i) all 3 students walk to school;

(2)

(ii) only Rita travels by bus to school;

(2)

(iii) at least 2 of the 3 students cycle to school.

(4)

(b) Ursula, a friend of Rita, never travels to school by bus. The probability that:

Ursula walks to school when Rita walks to school is 0.9;
Ursula cycles to school when Rita cycles to school is 0.7.

Calculate the probability that, on any given school-day morning, Rita and Ursula travel to school by:

(i) the same method;

(3)

(ii) different methods.

(1)
(Total 12 marks)

Q18.

Hugh owns a small farm.

(a) He has two sows, Josie and Rosie, which he feeds at a trough in their field at 8.00 am each day.

The probability that Josie is waiting at the trough at 8.00 am on any given day is 0.90.

The probability that Rosie is waiting at the trough at 8.00 am on any given day is 0.70 when Josie is waiting at the trough, but is only 0.20 when Josie is not waiting at the trough.

Calculate the probability that, at 8.00 am on a given day:

(i) both sows are waiting at the trough;

(2)

(ii) neither sow is waiting at the trough;

(2)

(iii) at least one sow is waiting at the trough.

(1)

(b) Hugh also has two cows, Daisy and Maisy. Each day at 4.00 pm, he collects them from the gate to their field and takes them to be milked.

The probability, P(D), that Daisy is waiting at the gate at 4.00 pm on any given day is 0.75.
The probability, P(M), that Maisy is waiting at the gate at 4.00 pm on any given day is 0.60.
The probability that both Daisy and Maisy are waiting at the gate at 4.00 pm on any given day is 0.40.

(i) In the table of probabilities, D' and M' denote the events 'not D' and 'not M' respectively.

	M	M'	Total
D	0.40		0.75
D'			
Total	0.60		1.00

Complete the table above.

(2)

(ii) Hence, or otherwise, find the probability that, at 4.00 pm on a given day:

(A) neither cow is waiting at the gate;

(1)

(B) only Daisy is waiting at the gate;

(1)

(C) exactly one cow is waiting at the gate.

(2)

(Total 11 marks)

Q19.

It is proposed to introduce, for all males at age 60, screening tests, A and B, for a certain disease.

Test B is administered only when the result of Test A is inconclusive.

It is known that 10% of 60-year-old men suffer from the disease.

For those 60-year-old men suffering from the disease:

- Test A is known to give a positive result, indicating a presence of the disease, in 90% of cases, a negative result in 2% of cases and a requirement for the administration of Test B in 8% of cases;
- Test B is known to give a positive result in 98% of cases and a negative result in 2% of cases.

For those 60-year-old men not suffering from the disease:

- Test A is known to give a positive result in 1% of cases, a negative result in 80% of cases and a requirement for the administration of Test B in 19% of cases;
- Test B is known to give a positive result in 1% of cases and a negative result in 99% of cases.

(a) Draw a tree diagram to represent the above information.

(4)

(b) (i) Hence, or otherwise, determine the probability that:

(A) a 60-year-old man, suffering from the disease, tests negative;

(B) a 60-year-old man, not suffering from the disease, tests positive.

(2)

(ii) A random sample of ten thousand 60-year-old men is given the screening tests. Calculate, to the nearest 10, the number who you would expect to be given an **incorrect** diagnosis.

(2)

(c) Determine the probability that:

(i) a 60-year-old man suffers from the disease given that the tests provide a positive result;

(ii) a 60-year-old man does not suffer from the disease given that the tests provide a negative result.

(5)
(Total 13 marks)

Q20.

A large bookcase contains two types of book: hardback and paperback. The number of books of each type in each of four subject categories is shown in the table.

		Subject category				
		Crime	Romanc e	Science fiction	Thriller	Total
Type	Hardback	8	16	18	18	60
	Paperback	16	40	14	30	100
	Total	24	56	32	48	160

(a) A book is selected at random from the bookcase. Calculate the probability that the book is:

(i) a paperback;

(1)

(ii) not science fiction;

(2)

(iii) science fiction or a hardback;

(2)

(iv) a thriller, given that it is a paperback.

(2)

(b) Three books are selected at random, without replacement, from the bookcase.

Calculate, to three decimal places, the probability that one is crime, one is romance and one is science fiction.

(4)

(Total 11 marks)

Q21.

A hotel chain has hotels in three types of location: city, coastal and country. The percentages of the chain's reservations for each of these locations are 30, 55 and 15 respectively.

Each of the chain's hotels offers three types of reservation: Bed & Breakfast, Half Board and Full Board.

The percentages of these types of reservation for **each** of the three types of location are shown in the table.

		Type of location		
		City	Coastal	Country
Type of reservation	Bed & Breakfast	80	10	30

	Half Board	15	65	50
	Full Board	5	25	20

For example, 80 per cent of reservations for hotels in city locations are for Bed & Breakfast.

(a) For a reservation selected at random:

(i) show that the probability that it is for Bed & Breakfast is 0.34;

(2)

(ii) calculate the probability that it is for Half Board in a hotel in a coastal location;

(2)

(iii) calculate the probability that it is for a hotel in a coastal location, given that it is for Half Board.

(4)

(b) A random sample of 3 reservations for Half Board is selected.

Calculate the probability that these 3 reservations are for hotels in different types of location.

(5)

(Total 13 marks)

Q22.

David and his partner, Amber, both work at their district hospital. The probability that David cycles to work each morning is 0.85. On a morning when David cycles to work, the probability that Amber cycles to work is 0.80. On a morning when David does not cycle to work, the probability that Amber cycles to work is 0.30.

(a) Calculate the probability that, on a particular morning:

(i) they both cycle to work;

(2)

(ii) only Amber cycles to work;

(2)

(iii) exactly one of them cycles to work.

(2)

(b) Assuming that decisions as to whether to cycle to work are independent from day to day, calculate the probability that, during a period of 4 working days, they both cycle to work on exactly 3 of the days and neither of them cycles to work on the other day.

(4)

(Total 10 marks)

Q23.

A health club has a number of facilities which include a gym and a sauna. Andrew and his wife, Heidi, visit the health club together on Tuesday evenings.

On any visit, Andrew uses either the gym or the sauna or both, but no other facilities. The probability that he uses the gym, $P(G)$, is 0.70. The probability that he uses the sauna, $P(S)$, is 0.55. The probability that he uses both the gym and the sauna is 0.25.

(a) Calculate the probability that, on a particular visit:

(i) he does not use the gym;

(1)

(ii) he uses the gym but not the sauna;

(2)

(iii) he uses either the gym or the sauna but not both.

(2)

(b) Assuming that Andrew's decision on what facility to use is independent from visit to visit, calculate the probability that, during a month in which there are exactly four Tuesdays, he does not use the gym.

(2)

(c) The probability that Heidi uses the gym when Andrew uses the gym is 0.6, but is only 0.1 when he does not use the gym.

Calculate the probability that, on a particular visit, Heidi uses the gym.

(3)

(d) On any visit, Heidi uses **exactly one** of the club's facilities.

The probability that she uses the sauna is 0.35.

Calculate the probability that, on a particular visit, she uses a facility other than the gym or the sauna.

(2)

(Total 12 marks)

Q24.

A basket in a stationery store contains a total of 400 marker and highlighter pens. Of the marker pens, some are permanent and the rest are non-permanent. The colours and types of pen are shown in the table.

	Colour			
Type	**Black**	**Blue**	**Red**	**Green**
Permanent marker	44	66	32	18
Non-permanent marker	36	53	21	10

Highlighter	0	41	37	42

A pen is selected at random from the basket. Calculate the probability that it is:

(a) a blue pen;

(1)

(b) a marker pen;

(2)

(c) a blue pen or a marker pen;

(2)

(d) a green pen, given that it is a highlighter pen;

(2)

(e) a non-permanent marker pen, given that it is a red pen.

(2)
(Total 9 marks)

Q25.

A manufacturer produces three models of washing machine: basic, standard and deluxe. An analysis of warranty records shows that 25% of faults are on basic machines, 60% are on standard machines and 15% are on deluxe machines.

For basic machines, 30% of faults reported during the warranty period are electrical, 50% are mechanical and 20% are water-related.

For standard machines, 40% of faults reported during the warranty period are electrical, 45% are mechanical and 15% are water-related.

For deluxe machines, 55% of faults reported during the warranty period are electrical, 35% are mechanical and 10% are water-related.

(a) Draw a tree diagram to represent the above information.

(3)

(b) Hence, or otherwise, determine the probability that a fault reported during the warranty period:

(i) is electrical;

(2)

(ii) is on a deluxe machine, given that it is electrical.

(2)

(c) A random sample of 10 electrical faults reported during the warranty period is selected. Calculate the probability that exactly 4 of them are on deluxe machines.

(3)
(Total 10 marks)

Q26.

The British and Irish Lions 2005 rugby squad contained 50 players. The nationalities and playing positions of these players are shown in the table.

		Nationality			
		English	**Welsh**	**Scottish**	**Irish**
Playing position	**Forward**	14	5	2	6
	Back	8	7	2	6

(a) A player was selected at random from the squad for a radio interview. Calculate the probability that the player was:

(i) English;

(2)

(ii) Irish, given that the player was a back;

(2)

(iii) a forward, given that the player was not Scottish.

(2)

(b) Four players were selected at random from the squad to visit a school. Calculate the probability that all four players were English.

(3)

(Total 9 marks)

Q27.

The British and Irish Lions 2005 rugby squad contained 50 players. The nationalities and playing positions of these players are shown in the table.

		Nationality			
		English	**Welsh**	**Scottish**	**Irish**
Playing position	**Forward**	14	5	2	6
	Back	8	7	2	6

(a) A player was selected at random from the squad for a radio interview. Calculate the probability that the player was:

(i) a Welsh back;

(1)

(ii) English;

(2)

(iii) not English;

(1)

(iv) Irish, given that the player was a back;

(2)

(v) a forward, given that the player was not Scottish.

(2)

(b) Four players were selected at random from the squad to visit a school. Calculate the probability that all four players were English.

(3)
(Total 11 marks)

Q28.

A hill-top monument can be visited by one of three routes: road, funicular railway or cable car. The percentages of visitors using these routes are 25, 35 and 40 respectively.

The age distribution, in percentages, of visitors using **each** route is shown in the table. For example, 15 per cent of visitors using the road were under 18.

		Percentage of visitors using		
		Road	**Funicular railway**	**Cable car**
Age (years)	**Under 18**	15	25	10
	18 to 64	80	60	55
	Over 64	5	15	35

Calculate the probability that a randomly selected visitor:

(a) who used the road is aged 18 or over;

(1)

(b) is aged between 18 and 64;

(3)

(c) used the funicular railway and is aged over 64;

(2)

(d) used the funicular railway, given that the visitor is aged over 64.

(5)
(Total 11 marks)

Q29.

Xavier, Yuri and Zara attend a sports centre for their judo club's practice sessions. The probabilities of them arriving late are, independently, 0.3, 0.4 and 0.2 respectively.

(a) Calculate the probability that for a particular practice session:

(i) all three arrive late;

(1)

(ii) none of the three arrives late;

(2)

(iii) only Zara arrives late.

(2)

(b) Zara's friend, Wei, also attends the club's practice sessions. The probability that Wei arrives late is 0.9 when Zara arrives late, and is 0.25 when Zara does not arrive late.

Calculate the probability that for a particular practice session:

(i) both Zara and Wei arrive late;

(2)

(ii) either Zara or Wei, but not both, arrives late.

(3)

(Total 10 marks)

Q30.

A housing estate consists of 320 houses: 120 detached and 200 semi-detached. The numbers of children living in these houses are shown in the table.

	Number of children				
	None	**One**	**Two**	**At least three**	**Total**
Detached house	24	32	41	23	120
Semi-detached house	40	37	88	35	200
Total	64	69	129	58	320

A house on the estate is selected at random.

D denotes the event 'the house is detached'.
R denotes the event 'no children live in the house'.
S denotes the event 'one child lives in the house'.
T denotes the event 'two children live in the house'.
(D' denotes the event 'not D'.)

(a) Find:

(i) P(D);

(1)

(ii) $P(D \cap R)$;

(1)

(iii) $P(D \mid R)$;

(2)

(iv) $P(R \mid D')$.

(3)

(b) (i) Name two of the events D, R, S and T that are mutually exclusive.

(1)

(ii) Determine whether the events D and R are independent. Justify your answer.

(2)

(c) Define, in the context of this question, the event:

(i) $D' \cup T$;

(2)

(ii) $D \cap (R \cup S)$.

(2)

(Total 14 marks)

Q31.

A housing estate consists of 320 houses: 120 detached and 200 semi-detached. The numbers of children living in these houses are shown in the table.

	Number of children				
	None	**One**	**Two**	**At least three**	**Total**
Detached house	24	32	41	23	120
Semi-detached house	40	37	88	35	200
Total	64	69	129	58	320

A house on the estate is selected at random.

D denotes the event 'the house is detached'.
R denotes the event 'no children live in the house'.
S denotes the event 'one child lives in the house'.
T denotes the event 'two children live in the house'.
(D' denotes the event 'not D'.)

(a) Find:

(i) P(D);

(1)

(ii) P($D \cap R$);

(1)

(iii) P($D \cup T$);

(2)

(iv) P($D \mid R$).

(2)

(v) P($R \mid D'$).

(3)

(b) (i) Name two of the events D, R, S and T that are mutually exclusive.

(1)

(ii) Determine whether the events D and R are independent. Justify your answer.

(2)

(c) Define, in the context of this question, the event:

(i) $D' \cup T$;

(2)

(ii) $D \cap (R \cup S)$.

(2)

(Total 16 marks)

Q32.

Each enquiry received by a business support unit is dealt with by Ewan, Fay or Gaby. The probabilities of them dealing with an enquiry are 0.2, 0.3 and 0.5 respectively.

Of enquiries dealt with by Ewan, 60% are answered immediately, 25% are answered later the same day and the remainder are answered at a later date.

Of enquiries dealt with by Fay, 75% are answered immediately, 15% are answered later the same day and the remainder are answered at a later date.

Of enquiries dealt with by Gaby, 90% are answered immediately and the remainder are answered at a later date.

(a) Determine the probability that an enquiry:

(i) is dealt with by Gaby and answered immediately;

(1)

(ii) is answered immediately;

(3)

(iii) is dealt with by Gaby, given that it is answered immediately.

(3)

(b) Determine the probability that an enquiry is dealt with by Ewan, given that it is answered later the same day.

(4)

(Total 11 marks)

Q33.

The table below shows the numbers of males and females in each of three employment categories at a university on 31 July 2003.

	Employment category		
	Managerial	**Academic**	**Support**
Male	38	369	303
Female	26	275	643

(a) An employee is selected at random. Determine the probability that the employee is:

(i) female;

(1)

(ii) a female academic;

(1)

(iii) either female or academic or both;

(2)

(iv) female, given that the employee is academic.

(2)

(b) Three employees are selected at random, without replacement. Determine the probability that:

(i) all three employees are male;

(2)

(ii) exactly one employee is male.

(3)

(c) The event “employee selected is academic” is denoted by A. The event “employee selected is female” is denoted by F.

Describe in context, as simply as possible, the events denoted by:

(i) $F \cap A$;

(1)

(ii) $F' \cup A$.

(2)

(Total 14 marks)

Q34.

A reliable estimate for the proportion of a population of fish with a certain disease is 60 per cent.

A test for the presence of the disease in a fish is possible. The test gives one of three conclusions: diseased, inconclusive, non-diseased.

For a **diseased** fish, the probabilities of these three conclusions are:

diseased	0.75
inconclusive	0.15
non-diseased	0.10

For a **non-diseased** fish, the probabilities of these three conclusions are:

diseased	0.05
inconclusive	0.15
non-diseased	0.80

(a) A fish is selected at random. Using a tree diagram, or otherwise, calculate the probability that:

(i) the fish has the disease and the test concludes that it is diseased;

(2)

(ii) the test concludes that the fish has the disease;

(3)

(iii) the test gives a correct conclusion.

(2)

(b) Three fish, all with the disease, are tested.

Find the probability that the test concludes that two fish are diseased and one fish is non-diseased.

(3)

(Total 10 marks)

Q35.

Fred and his daughter, Delia, support their town's rugby team. The probability that Fred watches a game is 0.8. The probability that Delia watches a game is 0.9 when her father watches the game, and is 0.4 when her father does not watch the game.

(a) Calculate the probability that:

(i) both Fred and Delia watch a particular game;

(2)

(ii) neither Fred nor Delia watch a particular game.

(2)

(b) Molly supports the same rugby team as Fred and Delia. The probability that Molly watches a game is 0.7, and is independent of whether or not Fred or Delia watches the game.

Calculate the probability that:

(i) all 3 supporters watch a particular game;

(3)

(ii) exactly 2 of the 3 supporters watch a particular game.

(4)

(Total 11 marks)